

Multi-Agent Social Media Company

Live Demo Record

A measured, fully local execution record — from clean-clone setup to fallback reproducibility and live-model behavior.

Execution	Fully local via Ollama; no hosted LLM APIs
Repository	github.com/yashieeeeeee/social_ai_company
Model	qwen2.5:3b · Q4_K_M · 1.9 GB · one model across 8 agents
Hardware	Intel i5-1240P · 8 GB RAM · CPU-only · Windows 64-bit
Date	2026-09-14

Purpose. This record documents the system exactly as it ran on the candidate machine: clean-clone setup, tests, local-model proof, a deterministic two-week baseline, and the same campaign driven by the live local model. Every number is measured, and every figure is a screenshot of the actual run.

1. What the system does

A human brief goes in. The Orchestrator parses it, Strategy plans a 12-post week across three mock channels (buzz, forum, pro), Writer and Creative draft each post, Compliance reviews with a 3-strike cap into a human queue, a CLI approval gate must say yes before anything publishes, the Scheduler publishes to a SQLite mock platform with hidden engagement rules R1-R7, the Community Manager answers comments or escalates, and Analytics produces a numeric weekly report whose recommendations feed Week 2. Full agent trace lands in logs/trace.jsonl.

2. Architecture and message flow

Agents never call each other. Every handoff is `bus.send()` into a shared SQLite messages table mirrored to `trace.jsonl`, routed by CampaignRunner.

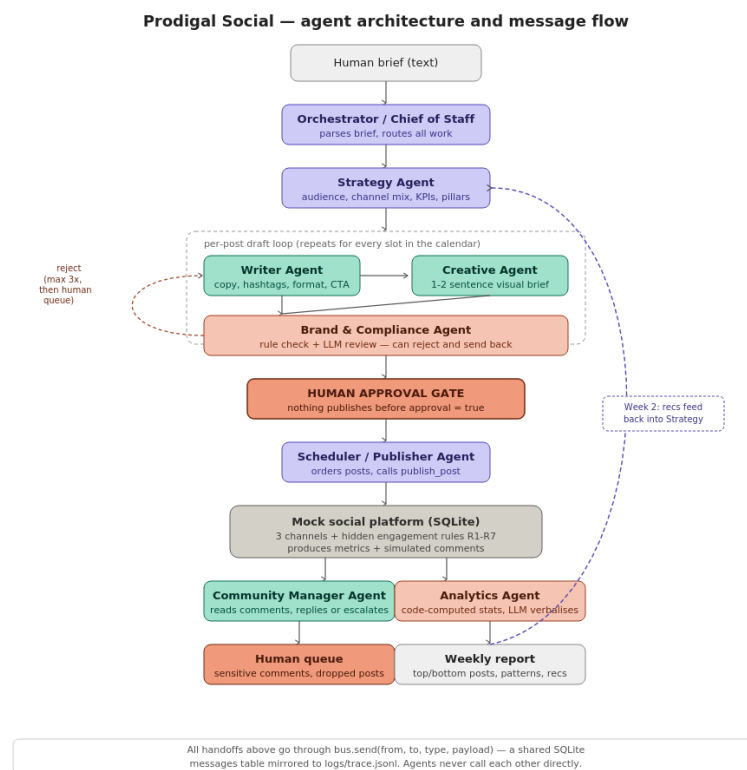


Figure 1. Prodigal Social agent architecture and message flow.

3. Setup from a clean clone

Cloned into an empty folder, installed one dependency (requests), ran with `--no-llm`. No model weights, services, or setup scripts needed; SQLite initialises itself on first run.

```

PS C:\Users\Acer> git clone https://github.com/yashieeeee/social_ai_company.git fresh-test-clone
Cloning into 'fresh-test-clone'...
remote: Enumerating objects: 65, done.
remote: Counting objects: 100% (65/65), done.
remote: Compressing objects: 100% (41/41), done.
remote: Total 65 (delta 23), reused 57 (delta 19), pack-reused 0 (from 0)
Receiving objects: 100% (65/65), 47.43 KiB | 1.39 MiB/s, done.
Resolving deltas: 100% (23/23), done.
PS C:\Users\Acer> cd fresh-test-clone
PS C:\Users\Acer\fresh-test-clone> pip install -r requirements.txt
Requirement already satisfied: requests in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from -r requirements.txt (line 1)) (2.34.2)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (3.19)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (2.7.0)
Requirement already satisfied: certifi=2023.5.7 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (2026.7.22)
[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Acer\fresh-test-clone> python run.py --brief-file demo_brief.txt --auto-approve --no-llm

```

Figure 2. Git clone and pip install in a fresh folder; proves zero-context setup.

4. Test suite

27 tests, mocked LLM (no model needed): hidden-rule spot checks, JSON repair and fallback behaviour, compliance retry cap, gate-fails-closed, streak detector, per-post error containment.

```

*
-----
Ran 27 tests in 1.084s
OK
PS C:\Users\Acer\fresh-test-clone> |

```

Figure 3. 27 tests ran in approximately one second on the fresh clone.

5. Local model proof

qwen2.5:3b pulled and listed locally (1.9 GB, sha256-verified). The only network call in the codebase is POST localhost:11434/api/generate.

```

OK
PS C:\Users\Acer\fresh-test-clone> ollama pull qwen2.5:3b
pulling manifest
pulling 5ee4f07cdb9b: 100%
pulling 66b9ea09bd5b: 100%
pulling eb4402837c78: 100%
pulling b5c0e5cf74cf: 100%
pulling 161ddde4c9cd: 100%
verifying sha256 digest
writing manifest
success
PS C:\Users\Acer\fresh-test-clone> ollama list
NAME      ID              SIZE    MODIFIED
qwen2.5:3b 357c53fb659c    1.9 GB  2 minutes ago
PS C:\Users\Acer\fresh-test-clone> |

```

Figure 4. Ollama pull and ollama list on the demo machine.

6. Demo run A — reproducible baseline, fallback mode

Same brief both weeks: budget espresso machine for students, playful, no over-promising. Week 1: 32,584 impressions, 1,427 comments, 3 escalations. Analytics found evening +106% reach (n=5/7) and x5.4 question-comment lift, so Week 2 shifted to evening slots: 42,812 impressions (delta +31%). Usage ledger 128 calls / 128 fallbacks: every output from deterministic templates, which is what makes it bit-for-bit reproducible for any evaluator without a GPU.

```

===== CAMPAIGN cmp-cli-1 (12 posts, round 1) =====
Objective: awareness | Audience: price-sensitive 18-25 students
[buzz d1 19:00] (howto/question) Student espresso for under a rupee a cup? price on a budget - what matters ... | coffee,students
[buzz d2 09:00] (story/none) Student espresso for under a rupee a cup. dorm-life on a budget, built for ... | coffee,students
[buzz d3 19:30] (poll/question) Student espresso for under a rupee a cup? taste on a budget - what matters ... | coffee,students
[buzz d5 03:00] (meme/none) Student espresso for under a rupee a cup. price on a budget, built for dorm... | coffee,students
[forum d1 12:30] (howto/question) Student espresso for under a rupee a cup? dorm-life on a budget - what matt... | coffee,students
[forum d2 19:00] (story/none) Student espresso for under a rupee a cup. taste on a budget, built for dorm... | coffee,students
[forum d4 09:00] (poll/question) Student espresso for under a rupee a cup? price on a budget - what matters ... | coffee,students
[forum d6 18:30] (meme/none) Student espresso for under a rupee a cup. dorm-life on a budget, built for ... | coffee,students
[pro d1 09:00] (howto/question) Student espresso for under a rupee a cup? taste on a budget - what matters ... | coffee
[pro d3 19:00] (story/none) Student espresso for under a rupee a cup. price on a budget, built for dorm... | coffee
[pro d4 12:30] (poll/question) Student espresso for under a rupee a cup? dorm-life on a budget - what matt... | coffee
[pro d6 08:30] (meme/none) Student espresso for under a rupee a cup. taste on a budget, built for dorm... | coffee
[gate] --auto-approve: approved
2026-09-14 16:34:48,456 prodigal.llm WARNING: LLM JSON call failed attempt 1 (orchestrator): --no-llm mode
2026-09-14 16:34:48,456 prodigal.llm WARNING: LLM JSON fallback (orchestrator): ollama-call-failed: --no-llm mode
[publish] 12 posts live.

```

Figure 5. Week-1 campaign: 12 posts, gate approved, 12 live.

```

[week1] imp=32584 comments=1427 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+106% imp)
REC: end 1 in 2 forum/buzz posts with a genuine question (x5.4 comments)

```

Figure 6. Week-1 report lines with the two numeric recommendations.

```

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *           approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *           approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *           approve
LLM usage: {'calls': 128, 'tokens_in': 0, 'tokens_out': 0, 'fallbacks': 128}

```

Figure 7. Week 2 delta +31%, bus handoffs, and usage ledger.

7. Demo run B — live local model, the headline

Same pipeline, model doing the writing: 134 calls, 26,310 tokens in, 8,885 out, 0 fallbacks. The trace caught a live compliance reject-then-rewrite-then-approve on camera. Week 1: 21,167 impressions, 542 comments. Week 2: 16,583 (delta -22%). The regression is understood, not hidden: model drafts collapsed to one format (11/12 howto) with zero question-endings, so Week 2 kept the evening gain but ate the R5 novelty decay penalty while harvesting none of R2. A second interactive run (human typed y at the gate) gave 149 calls, 2 fallbacks (analytics timeouts), W1 20,322 vs W2 16,043 (delta -21%). Honest numbers beat flattering ones; the loop works, the model has a diversity failure with a named mechanism.

```
Windows PowerShell
[publish] 12 posts live.
[week1] imp=21167 comments=542 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+131% imp)
REC: end 1 in 2 forum/buzz posts with a genuine question (x1.9 comments)
[week2] imp=16583 (delta -22%)

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> * approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[reject]--> writer
orchestrator --[compliance_attempt]--> * reject
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> * approve

LLM usage: {'calls': 134, 'tokens_in': 26310, 'tokens_out': 8885, 'fallbacks': 0}
PS D:\dev\prodigal social>
```

Figure 8. Live-model campaign: 11 posts, Week 1/Week 2 lines, 134 calls, 0 fallbacks.

```
Windows PowerShell
rs_BehindTheScenes
[forum dt 09:00] (poll/Share your favorite espresso story in the comments!) Did you know espresso was first brewed in Italy in the 16th century? It's n... | espresso,coffeeun,italiancult
ture
[forum dt 10:30] (name/Give it a shot and let me know how it turns out!) Hey coffee lovers! Ever wished your morning cup was as perfect as that fanc... | coffeebrewing,morningcoffee,coff
eeaddicts
[pro dt 09:00] (howto/Watch now to learn more about your favorite brew!) Ever wondered how your perfect cup of espresso is made? Today at 9 AM, join... | espresso,coffeemaking,barista
[pro dt 10:00] (howto/What's your favorite way to enjoy espresso?) Did you know? Espresso is named after espresso coffee, which is a quick way... | FunFacts,EspressoLovers,CoffeeCult
ure
[pro dt 12:30] (howto/Try it out and let me know how it goes!) Got a coffee craving? Here's how to make the perfect cup at home. Start wit... | coffee,brewing,homebrew
[pro dt 08:30] (howto/Learn More) Ever wondered how your favorite café keeps its espresso shots so perfect? L... | espresso,coffeemachine,barista
[y]approve [s]re-plan strategy [w]rewrite posts [N]quit > y

[publish] 12 posts live.
2026-09-14 18:08:10.743 prodigal.llm WARNING: LLM JSON call failed attempt 1 (analytics): HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
2026-09-14 18:08:10.831 prodigal.llm WARNING: LLM JSON fallback (analytics): ollama-call-failed: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
[week1] imp=20322 comments=516 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+123% imp)
2026-09-14 18:25:42.816 prodigal.llm WARNING: LLM JSON call failed attempt 1 (analytics): HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
2026-09-14 18:25:43.159 prodigal.llm WARNING: LLM JSON fallback (analytics): ollama-call-failed: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
[week2] imp=16043 (delta -21%)

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[reject]--> writer
orchestrator --[compliance_attempt]--> * reject
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> * approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> * approve

LLM usage: {'calls': 149, 'tokens_in': 28807, 'tokens_out': 8340, 'fallbacks': 2}
PS C:\Users\Acer\fresh-test-clone>
```

Figure 9. Interactive gate with typed y, reject rows in trace, 149 calls, and 2 fallbacks.

8. Reproduce everything

```
git clone https://github.com/yashieeeeeee/social_ai_company.git
pip install -r requirements.txt
python -m unittest discover -s tests
python run.py --brief-file demo_brief.txt --auto-approve --no-llm
python run.py --brief-file demo_brief.txt --auto-approve # needs Ollama
```

9. Honest scorecard vs planted rules R1-R7

Found: R1 prime-time (acted on, measured lift), R2 question boost (needs draft variance). Noted: R6 CTA effect. Missed by design: R3/R4/R7 are censored by Compliance before publish (rejected drafts logged as counterfactual post_dropped messages); R5 needs a 3-day same-format streak the rotating calendar never forms (detector built and unit-tested on synthetic streaks). With the live model, Week 2 regressed via format collapse into R5 — documented above with the mechanism, not excused.